

Technology Stack

Date	06 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>Streamlit (Python)</i>	<i>Provides a simple, interactive web interface for students to access emotion detection, select subjects, and receive personalized learning recommendations.</i>

2	Backend / Server-Side	<i>Python, TensorFlow/ Keras, OpenCV</i>	<i>Python powers the application logic, TensorFlow/Keras performs emotion classification, and OpenCV captures and processes facial images from the webcam.</i>
3	Database / Data Storage	<i>CSV Files / Local Storage (or SQLite if you used it)</i>	<i>Stores emotion detection results, learning recommendations, and user progress efficiently for the prototype application.</i>
4	Cloud / Hosting / Deployment	<i>Streamlit Community Cloud</i>	<i>Enables easy deployment of the application online, allowing users to access it through a web browser without local installation.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Manages source code, version history, and collaboration among team members while supporting deployment from the repository.</i>

6	Third-Party APIs / Other Tools	<i>OpenCV, TensorFlow, NumPy, Pandas, Matplotlib</i>	<i>OpenCV captures webcam input, TensorFlow performs emotion detection, NumPy handles numerical operations, Pandas manages datasets, and Matplotlib visualizes results where required.</i>